

Introduction

Formal Properties in asynchronous computation:

- **Safety:** Nothing bad happens ever. If it is violated, it is done by a finite computation.

This is a safety property as ... violates the property and can be done in finite time. The “bad thing” is

- **Liveness:** Something good happens eventually. Cannot be violated by a finite computation (intuition: we can always run longer and see what happens).

This can be read as “... will eventually ...”. This is a liveness property as one cannot violate it in a finite number of steps: If ... yet, they may do so eventually at a later time. The “good thing” is